

ZIYI ZHU

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EXPERIENCE

Slingshot AI

Member of Technical Staff

New York, United States

Nov. 2024 – Present

- Led full end-to-end development of a foundation model on psychology, including **continual pre-training** on human therapy transcripts, **supervised fine-tuning** on golden conversations curated by clinical teams, and **alignment** on annotator-labeled preference datasets via offline and online policy optimization algorithms such as DPO and GRPO.
- Engineered scalable training infrastructure with hands-on experience in data preparation, cleaning, and multi-GPU distributed training with TRL and verl; scaled single-node training to **16k context** on **122B-parameter** models using PEFT techniques.
- Deployed language models and reward models (token and sequence classification) with vLLM and TensorRT with autoscaling in a managed Kubernetes cluster, serving both production traffic and development workloads.
- Developed value models to predict long-term outcomes and user behavioral signals, applying multi-objective RL optimization to **improve D1–D7 retention by ~3%** while increasing positive behaviors and therapeutic-process markers in production.
- Developed novel training methods, including synthetic preference datasets selected by learned value functions, and adversarial training for realistic user simulation, enabling offline evaluation and multi-turn policy optimization.
- Led design and development of an agentic prompt optimization system in which agents perform diagnostic code execution and propose targeted prompt changes, driving a **14% D1 retention improvement** over the quarter.
- Led initiatives to improve voice turn-taking experience; trained an audio-based endpointing model built on Whisper to detect end of turn, **reducing interruption rates by 70%** at equivalent latency targets.
- Developed internal tooling for annotation workflows and benchmark creation using TypeScript and Kotlin.

Unlikely AI

Senior Applied Scientist

London, United Kingdom

Jan. 2023 – Nov. 2024

- Architected core components of a neurosymbolic AI platform combining LLMs with symbolic reasoning, prioritizing accuracy, trustworthiness, and explainability for high-stakes decision-making in regulated industries.
- Engineered semantic parsing and entity resolution models, achieving **50% improvement** over open-source alternatives through transformer fine-tuning on synthetic datasets for insurance and financial services applications.
- Redesigned the symbolic language by integrating LLMs with symbolic solvers, delivering **99% precision** on tasks with **90% cost reduction**; this technology formed the foundation of the platform sold to Lloyds Banking Group.
- Achieved significant reduction in reasoning complexity through optimization of ontological reasoning using imitation learning and reinforcement learning techniques, enabling real-time inference at scale.

Chai Research

Founding AI Researcher

Palo Alto, United States

Dec. 2021 – Nov. 2022

- Built a social AI platform from inception to achieving **200k+ daily active users** and **1M+ daily messages**, establishing one of the largest real-world LLM deployments at the time.
- Achieved **60% D1 retention** through LLM fine-tuning and prompt engineering on curated conversational datasets, along with user-centric research and implementation of **20+ app features** in Flutter.
- Implemented encoder-based reward models trained on human preference data using rejection sampling, which **doubled user engagement** as measured by median conversation length.
- Designed, built, and deployed an offline reinforcement learning pipeline using PPO to maximize user engagement, marking one of the earliest successful applications of RLHF to social AI products at scale.
- Orchestrated the large-scale deployment of LLMs across CoreWeave GPU clusters, building high-throughput inference infrastructure that successfully served hundreds of models on thousands of GPUs with high availability.

Seamless Capital

Quantitative Researcher

Cambridge, United Kingdom

Aug. 2021 – Jan. 2022

- Developed machine learning models (GBDT, MLP, transformers, and ensemble methods) for multi-market horse racing prediction, including Win/Place/Quinella, deployed in live automated trading on Betfair exchange.
- Engineered **50+ alpha signals** and specialized features through historical data analysis, time-series feature extraction, and odds-implied probability calibration, resulting in **10% P&L improvement**.
- Implemented back-testing infrastructure with cross-validation, walk-forward optimization, and Monte Carlo simulation for risk analysis, validated through Sharpe ratio, drawdown metrics, and live market execution performance.

UBS Group AG

Financing Quant Intern

London, United Kingdom

June 2021 – Aug. 2021

- Optimized Stock Borrow Loan (SBL) desk operations through automated trade booking, position reconciliation, and settlement.
- Developed analytical tools on the Advanced Cloud Quantitative Analytics (ACQA) platform for repo rate curve construction, collateral valuation, and securities pricing models.

PUBLICATIONS

- **Z. Zhu**, D. R. Cahn, T. D. Hull, C. A. Stamatis, O. Tieleman, G. B. Freire, J. Chen. “Optimizing Sparse Outcomes Through Dense Behavioral Signals via Value-Guided Preference Distillation.” *Preprint*, 2026. [[arXiv:2609.14648](#)]
- **Z. Zhu**, O. Tieleman, A. Bukhtiyarov, J. Chen. “CyclicJudge: Mitigating Judge Bias Efficiently in LLM-based Evaluation.” *EMNLP 2026 (Main Conference)*. [[arXiv:2603.01865](#)]
- **Z. Zhu**, O. Tieleman, C. A. Stamatis, L. Smyth, T. D. Hull, D. R. Cahn, J. Chen, M. Malgaroli. “DIAL: Direct Iterative Adversarial Learning for Realistic Multi-Turn Dialogue Simulation.” *Findings of EMNLP 2026*. [[arXiv:2512.20773](#)]
- **Z. Zhu**, L. Smyth, S. Shinoda, J. Chen. “SAGE: Stochastic Prompt Optimization via Agent-Guided Exploration.” *EMNLP 2026 (Industry Track)*. [[arXiv:2606.18902](#)]
- R. Irvine, D. Boubert, V. Raina, A. Liusie, **Z. Zhu**, et al. “Rewarding Chatbots for Real-World Engagement with Millions of Users.” *Preprint*, 2023. [[arXiv:2303.06135](#)]

EDUCATION

University of Cambridge, Christ’s College

Cambridge, United Kingdom

Master of Arts (Cantab.)

May 2025

Bachelor of Arts, Engineering

Oct. 2018 – June 2021

- Results: First Class Honours, ranking in the **top 4% (12th out of 288 students)**, securing the Christ’s College Scholarship for **three consecutive years**.
- Modules: Information and Computer Engineering, Electrical and Information Sciences, and Instrumentation and Control.
- Achievements: Christ’s College Art Prize 2021, CUES Hackathon Champion 2021, VOMG Most Promising Musician 2017, and dual Silver Awards in Singapore Physics Olympiad (2014, 2016).

ACTIVITIES

Smartathon (Smart City Challenge) by Saudi Data & AI Authority (SDAIA)

Riyadh, Saudi Arabia

Second Prize Winner

Apr. 2023

- Secured **2nd place** among **14,731** teams globally, earning **SAR 97,500** in prize money and presenting the solution at Riyadh.
- Developed a pothole severity classification system combining YOLOv5 for real-time 2D object detection and DiffNet for monocular depth estimation and 3D surface reconstruction, enabling automated road maintenance optimization.

Personal Projects

- Engineered and deployed a profitable tennis match prediction system on Betfair, consistently achieving **~5%** daily returns.
- Derived closed-form probability model for stop-limit order execution using Optional Stopping Theorem and exponential martingales on geometric Brownian motion, validated through Monte Carlo simulations, achieving **<1%** relative error.

SKILLS & INTERESTS

Languages: Chinese (Native), English (Fluent), Japanese (Beginner)

Programming: Python, TypeScript, JavaScript, Kotlin, Java, C++, SQL, Flutter, React, MATLAB

LLM Training & Inference: PyTorch, Hugging Face Transformers, Accelerate, TRL, verl, DeepSpeed, FSDP, FlashAttention, PEFT techniques (LoRA, QLoRA), vLLM, TensorRT

Domains: Natural Language Processing, Computer Vision, Diffusion Models, Audio Processing (speech endpointing & generation)

Classical ML & Vision: XGBoost, LightGBM, OpenCV, scikit-learn, image segmentation, depth estimation

Infrastructure: Distributed training (multi-node, multi-GPU), Docker, Kubernetes, GCP, AWS, Weights & Biases, Nebius Cloud

Hobbies: Percussive fingerstyle guitar, piano, drum set, visual art, oil painting, photography, pool, tennis, table tennis, and travel